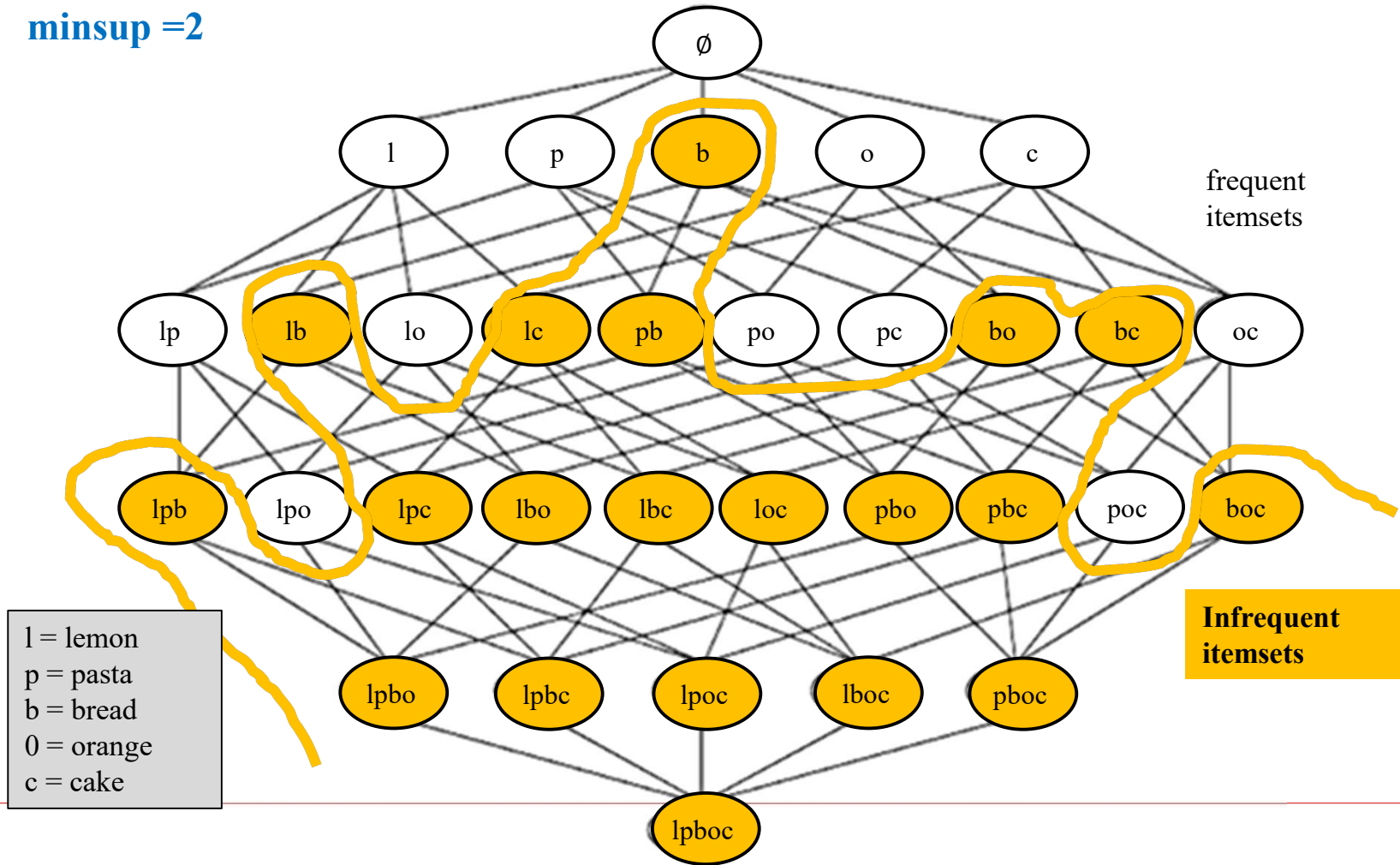

Rare pattern mining

We could define rare itemsets as **infrequent itemsets**

Definition 1 (infrequent itemset): An itemset X is infrequent if $\text{sup}(X) < \text{minsup}$.

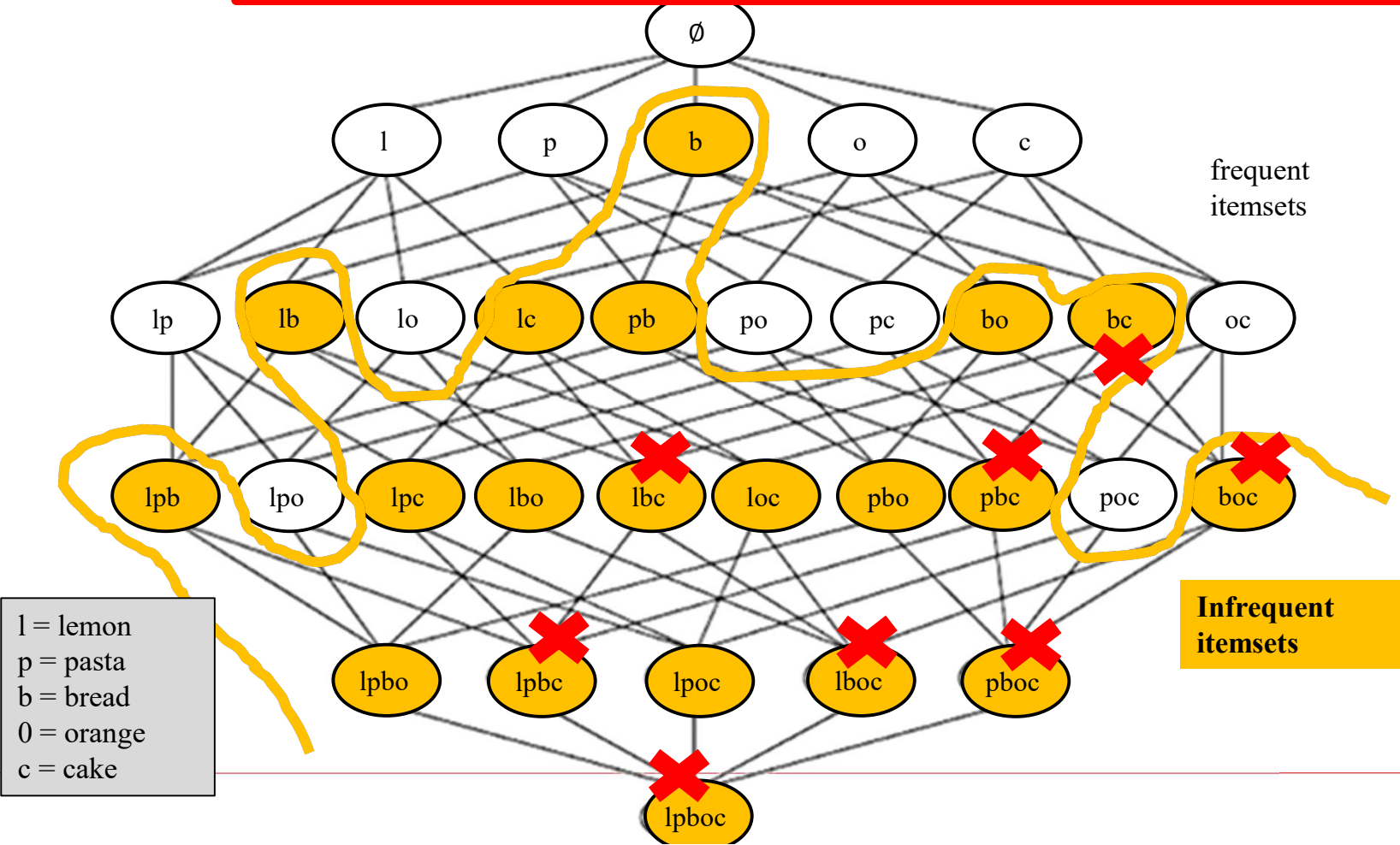
minsup = 2



We could define rare itemsets as **infrequent itemsets**

minsup = 2

Problem: Too many itemsets.
Some infrequent itemsets do not even exist (support = 0)



Another definition: minimal rare itemsets

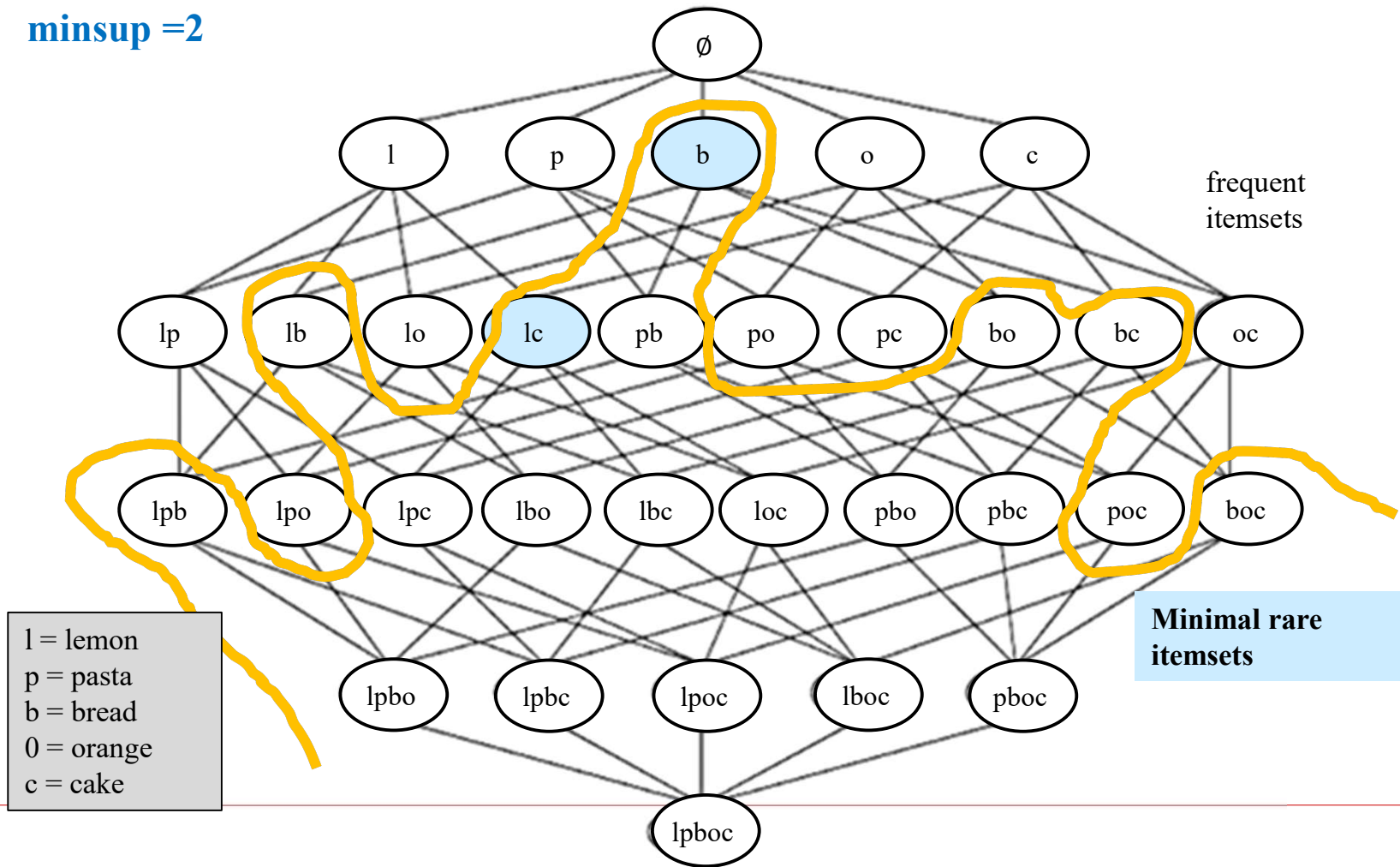
Proposed for the AprioriRare algorithm:

Laszlo Szathmary, Amedeo Napoli, Petko Valtchev: [Towards Rare Itemset Mining](#). ICTAI (1) 2007: 305-312

Definition 1 (minimal rare itemset): An itemset X is a minimal rare itemset if $\text{sup}(X) < \text{minsup}$ and all its proper subsets are frequent itemsets (i.e. for any subset $Y \subset X$, $\text{sup}(Y) \geq \text{minsup}$)

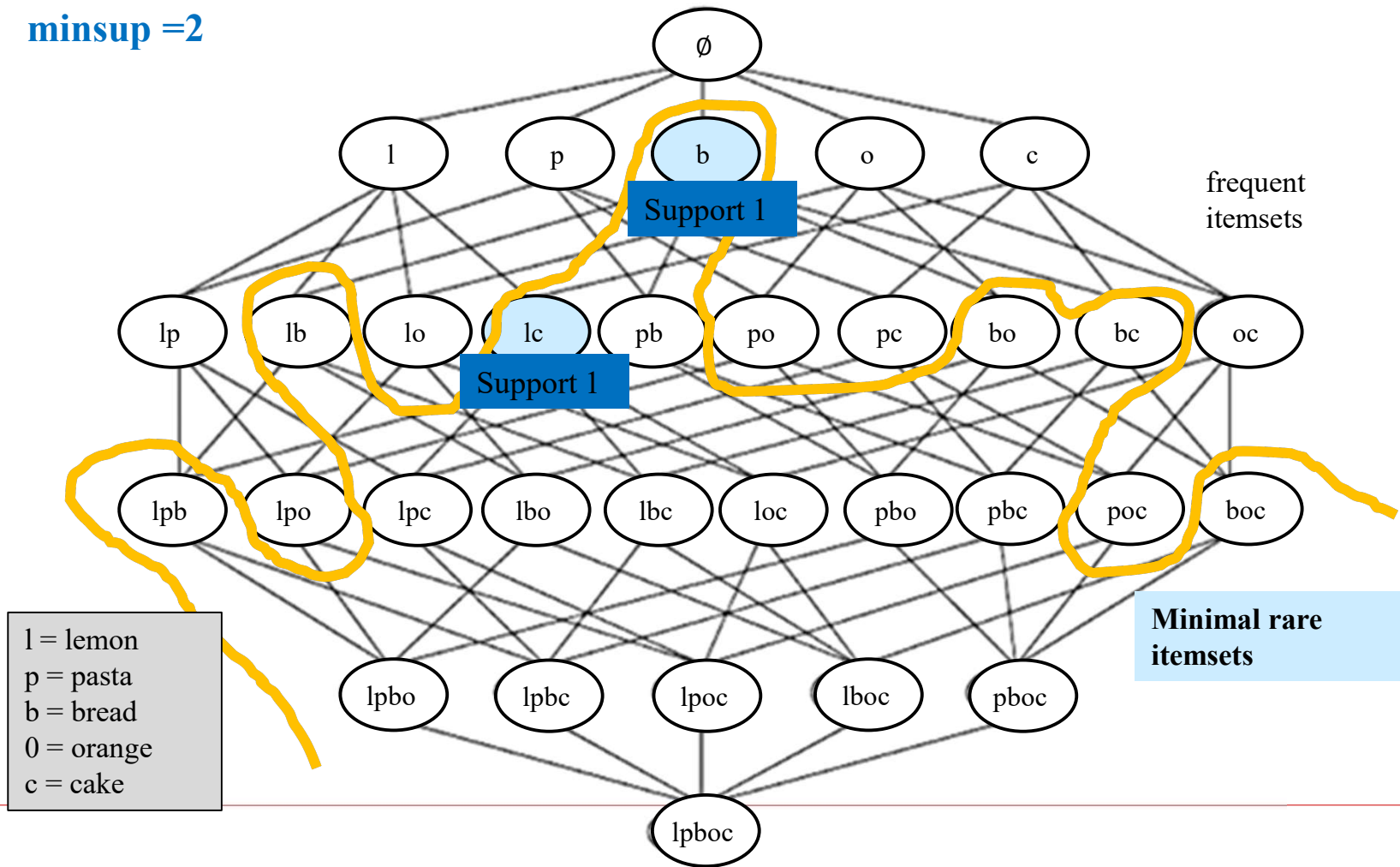
Minimal rare itemsets

minsup = 2



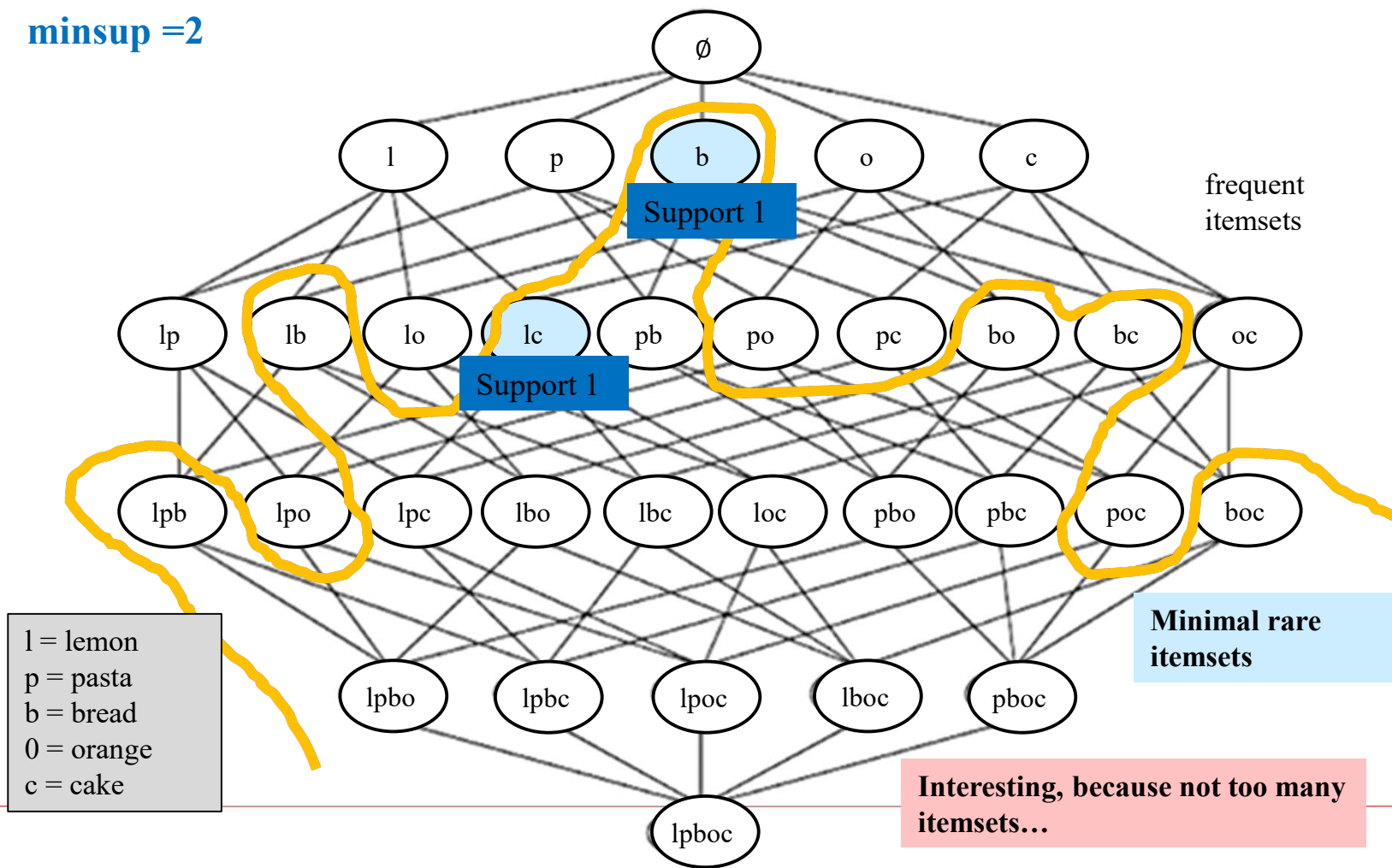
Minimal rare itemsets

minsup = 2



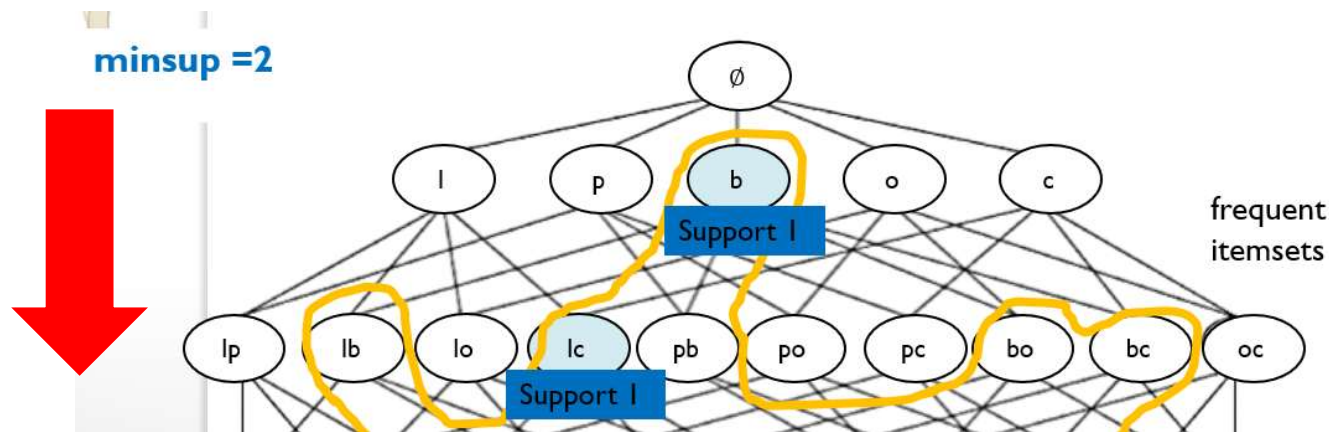
Minimal rare itemsets

minsup = 2



How can we find the minimal rare itemsets?

- ❑ It is not easy!
- ❑ Generally, frequent itemset mining algorithms start from single items and combine them to find larger itemsets.
- ❑ As itemsets become larger, the support can decrease.
- ❑ Thus, to search for rare itemsets, we must « pass through » the frequent itemsets to reach the rare itemsets. How?



The AprioriRare algorithm (2007)

- It is based on Apriori
- As Apriori, the itemsets are generated by levels:
 - Itemsets of size 1 (one item)
 - Itemsets of size 2 (two items)
 - Itemsets of size 3 (three items)
 - ...
- Two differences:
 - If AprioriRare finds an itemset of size k that is infrequent, AprioriRare checks if its subsets are frequent. If yes, it is a minimal rare itemset.
 - AprioriRare do not use the infrequent itemsets to generate larger itemsets.

The AprioriRare algorithm

- **Input:**
 - *minsup*
 - a transactional database
- **Output:**
 - all the minimal rare itemsets

Consider *minsup* = 2.

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

The AprioriRare algorithm

Step 5: For each infrequent itemset, check if all the subsets are frequent

Candidates of size 2

{pasta, lemon}	support: 3
{pasta, orange}	support: 3
{pasta, cake}	support: 2
{lemon, orange}	support: 2
{lemon, cake}	support: 1
{orange, cake}	support: 2

Minimal Rare
Itemset

Another definition: perfectly rare itemsets

*Proposed for the **AprioriInverse** algorithm:*

Yun Sing Koh, Nathan Rountree: [Finding Sporadic Rules Using Apriori-Inverse](#). PAKDD 2005: 97-106

Definition 1 (perfectly rare itemset):

Let there be two thresholds minsup and maxsup , such that $\text{maxsup} > \text{minsup}$.

An itemset Z is a **frequent itemset** if $\text{sup}(Z) \geq \text{maxsup}$.

An itemset X is a **perfectly rare itemset** if $\text{sup}(X) \geq \text{minsup}$ and $\text{sup}(X) < \text{maxsup}$ and for any non empty subset $Y \subset X$, $\text{sup}(Y) < \text{maxsup}$.

Definition 1 (perfectly rare itemset):

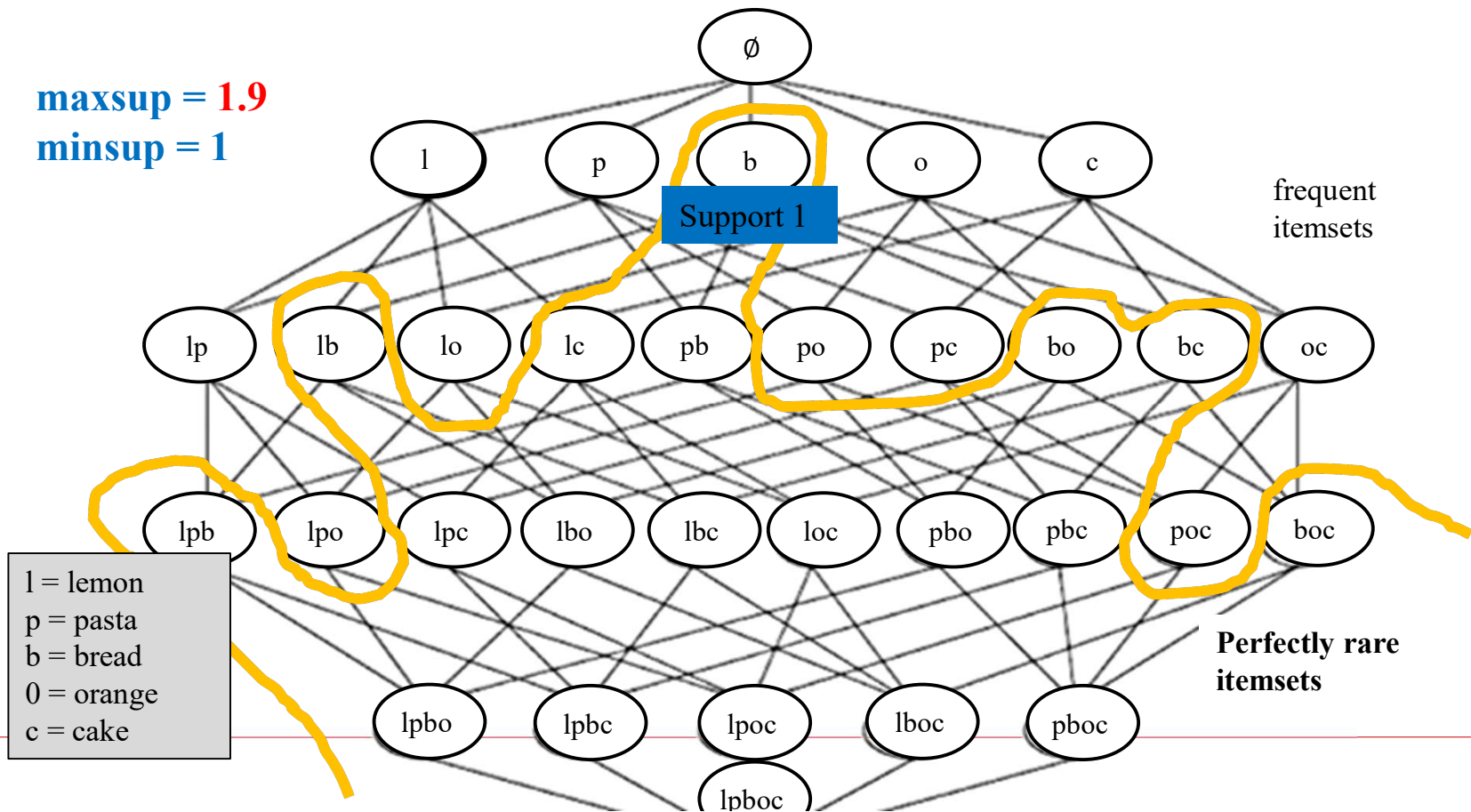
Let there be two thresholds minsup and maxsup , such that $\text{maxsup} > \text{minsup}$.

An itemset Z is a **frequent itemset** if $\text{sup}(Z) \geq \text{maxsup}$.

An itemset X is a **perfectly rare itemset** if $\text{sup}(X) \geq \text{minsup}$, $\text{sup}(X) < \text{maxsup}$ and for any non empty subset $Y \subset X$, $\text{sup}(Y) < \text{maxsup}$.

$\text{maxsup} = 1.9$

$\text{minsup} = 1$



Definition 1 (perfectly rare itemset):

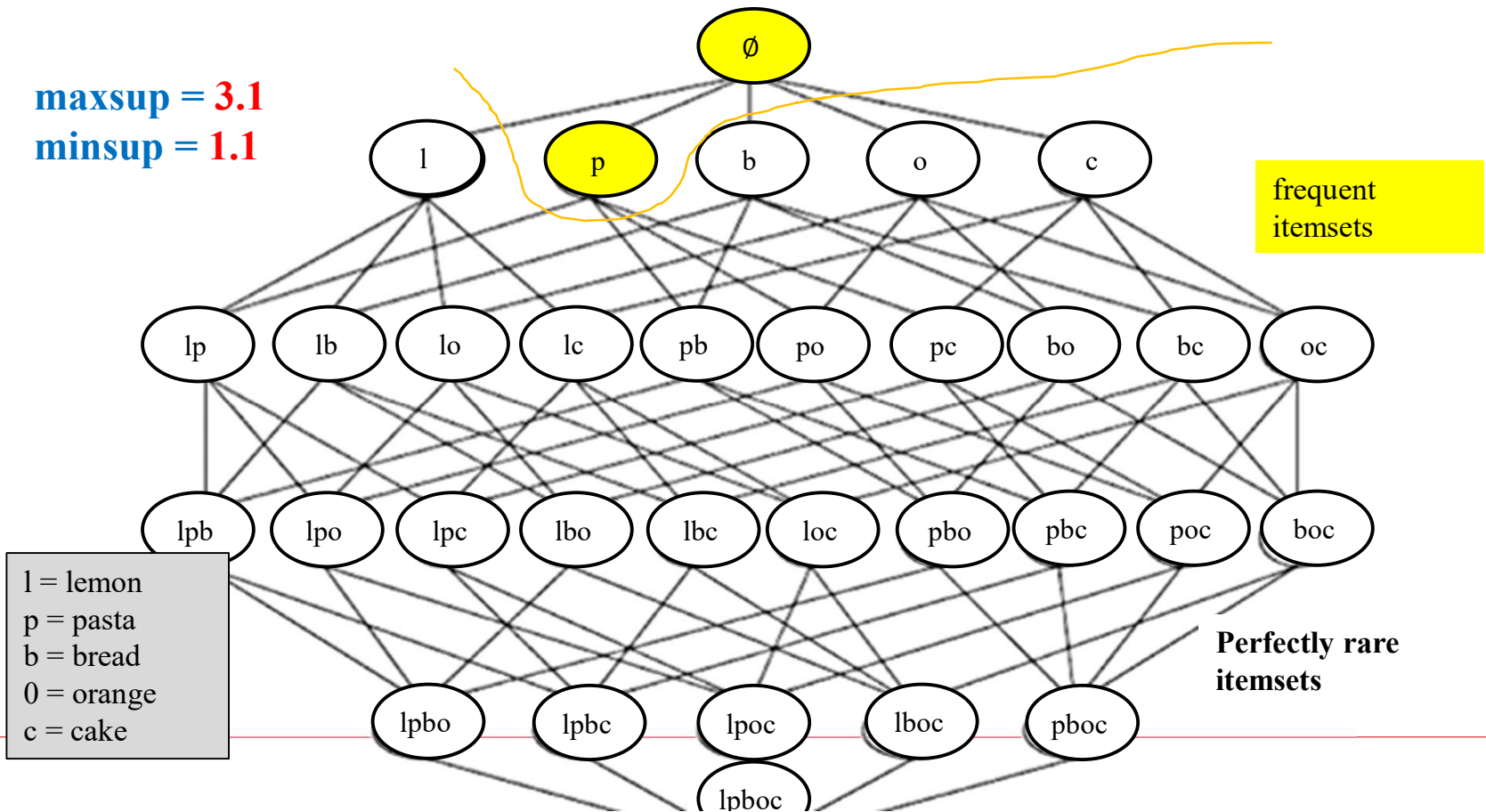
Let there be two thresholds minsup and maxsup , such that $\text{maxsup} > \text{minsup}$.

An itemset Z is a **frequent itemset** if $\text{sup}(Z) \geq \text{maxsup}$.

An itemset X is a **perfectly rare itemset** if $\text{sup}(X) \geq \text{minsup}$, $\text{sup}(X) < \text{maxsup}$ and for any non empty subset $Y \subset X$, $\text{sup}(Y) < \text{maxsup}$.

$\text{maxsup} = 3.1$

$\text{minsup} = 1.1$

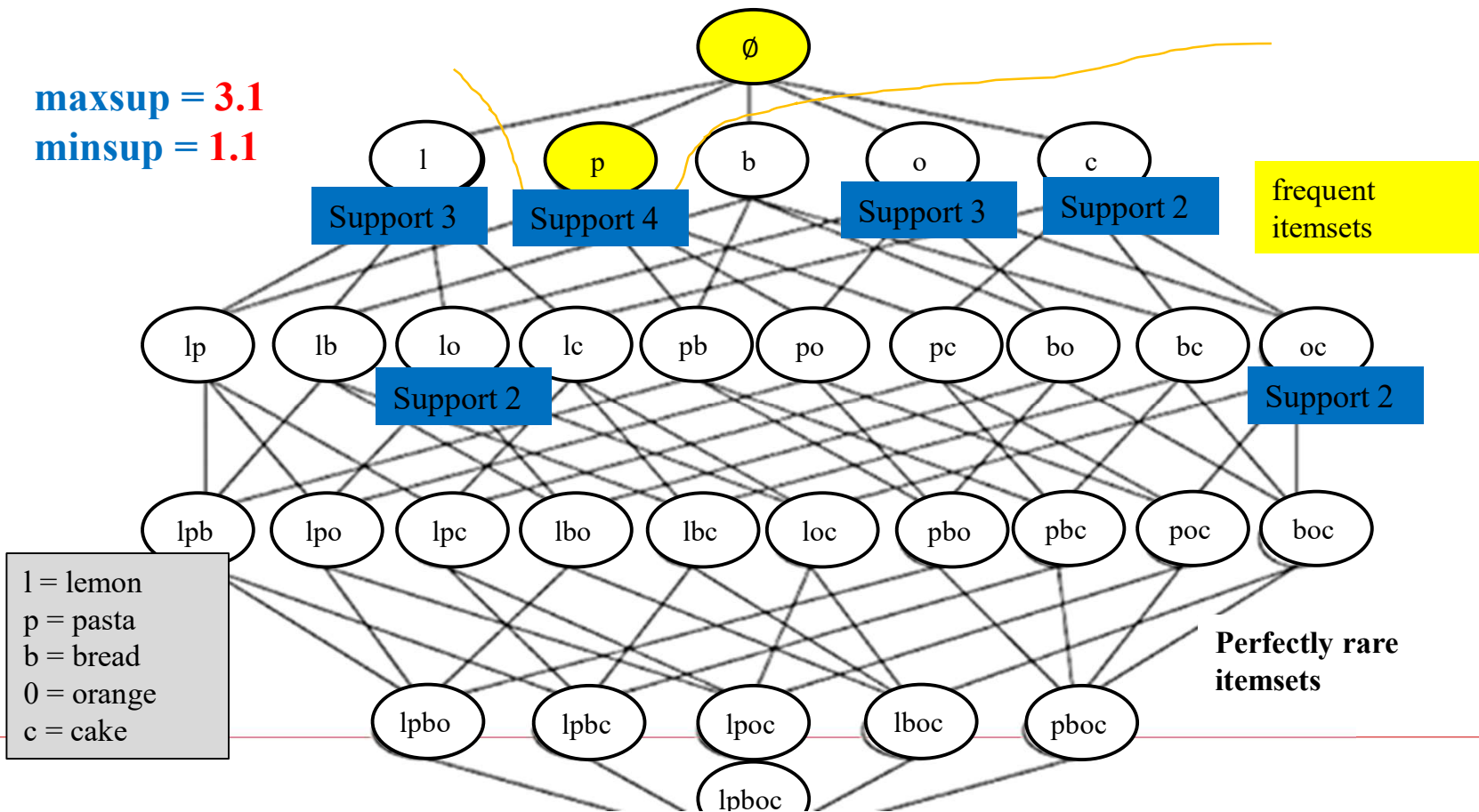


Definition 1 (perfectly rare itemset):

Let there be two thresholds `minsup` and `maxsup`, such that `maxsup > minsup`.

An itemset Z is a **frequent itemset** if $\text{sup}(Z) \geq \text{maxsup}$.

An itemset X is a **perfectly rare itemset** if $\text{sup}(X) \geq \text{minsup}$, $\text{sup}(X) < \text{maxsup}$ and for any non empty subset $Y \subset X$, $\text{sup}(Y) < \text{maxsup}$.



Definition 1 (perfectly rare itemset):

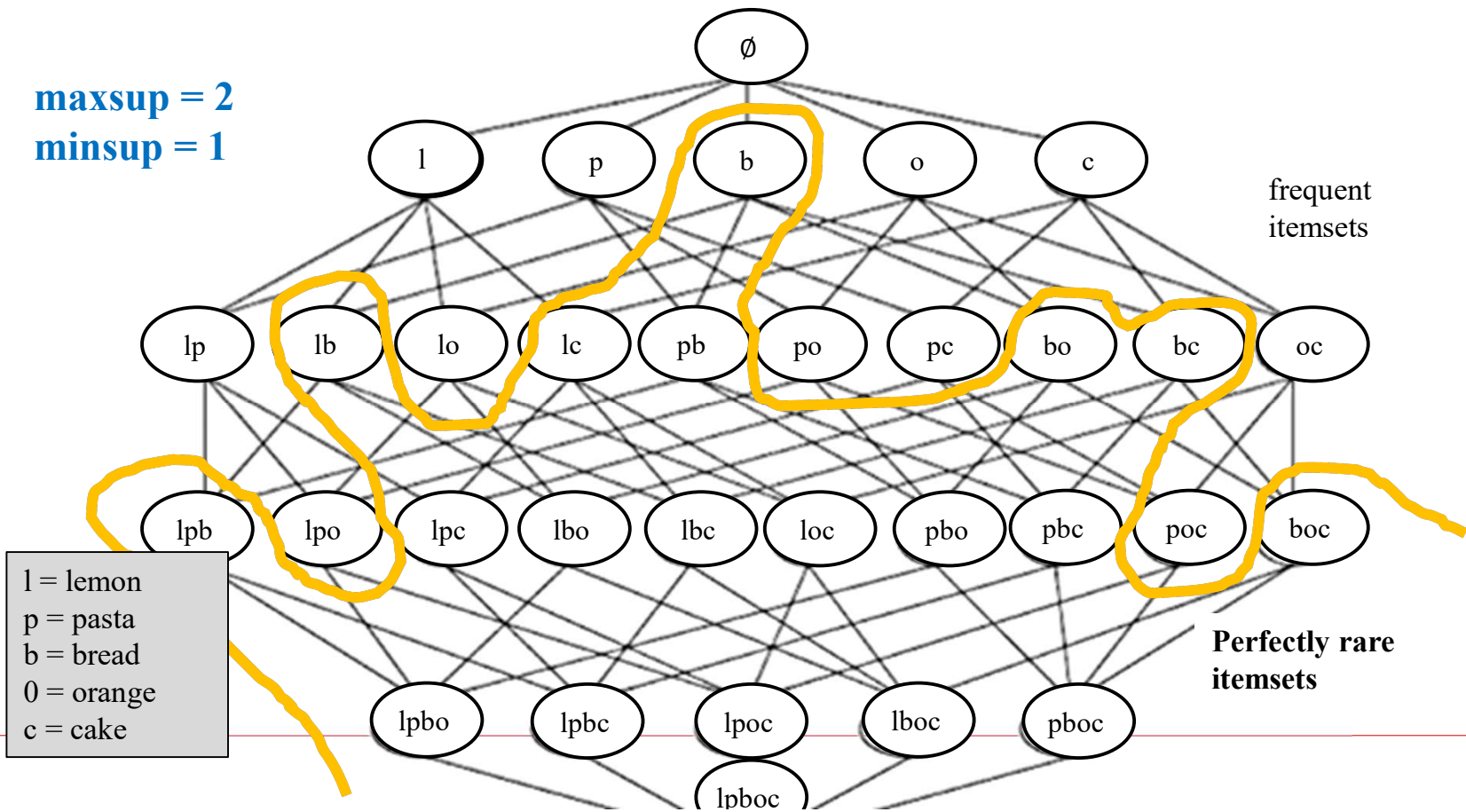
Let there be two thresholds minsup and maxsup , such that $\text{maxsup} > \text{minsup}$.

An itemset Z is a **frequent itemset** if $\text{sup}(Z) \geq \text{maxsup}$.

An itemset X is a **perfectly rare itemset** if $\text{sup}(X) \geq \text{minsup}$, $\text{sup}(X) < \text{maxsup}$ and for any non empty subset $Y \subset X$, $\text{sup}(Y) < \text{maxsup}$.

$\text{maxsup} = 2$

$\text{minsup} = 1$



How to find perfectly rare itemsets?

- ❑ AprioriInverse (2005)
- ❑ Based on Apriori.
- ❑ Key difference:
 - Initially, AprioriInverse discards each item x such that $\text{sup}(x) \geq \text{maxsup}$ because we don't need such item.
 - After that AprioriInverse search for itemsets using the remaining items just like Apriori to find itemsets with a support no less than *minsup*.

Correlated patterns mining

Frequent itemset mining

- Let there be a positive number *minsup*, set by the user.
- **Frequent itemset mining** (FIM) consists of enumerating all **frequent itemsets**, that is sets of values having a support greater or equal to *minsup* in a database *D*.

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange, cake}

Example

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

For *minsup* = 2, the frequent itemsets are:

{lemon}, {pasta}, {orange}, {cake}, {lemon, pasta}, {lemon, orange},
{pasta, orange}, {pasta, cake}, {orange, cake}, {lemon, pasta, orange}

These itemsets appear many times but **should we “believe” them?**

No. Many itemsets are spurious!

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

- ❑ A frequent itemset may contain items that are **weakly correlated (appear together more or less by chance)**.
 - ❑ e.g {pasta, cake} appears in 50% of the transactions.
 - ❑ But it does not mean much because pasta appears in all transactions!
-

How to solve this problem?

- ❑ **Solution 1:** Evaluate the correlation between items in an itemset using functions such as:
 - Bond
 - All-confidence
 - ❑ **Solution 2:** Apply statistical testing
 - ❑ **Solution 3:** Find other types of patterns, e.g., association rules.
-

The BOND Measure

Omiecinski, E.R.: Alternative interest measures for mining associations in databases. IEEE Transactions on Knowledge and Data Engineering 15(1), pp. 57-69 (2003)

(Conjunctive) support

The (conjunctive) support of an itemset X is the number of transactions that contains X .

$$\text{sup}(X) = |\{T \mid X \subseteq T \wedge T \in D\}|$$

Example: $\text{sup}(\{\text{pasta}, \text{orange}\}) = 3$

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange, cake}

Disjunctive support

The **disjunctive support of an itemset X** is the number of transactions that contains at least an item from X .

$$dsup(X) = |\{T \mid X \cap T \neq \emptyset \wedge T \in D\}|$$

Example: $dsup(\{\text{pasta}, \text{orange}\}) = 4$

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange, cake}

Bond

The **bond of an itemset X** is its conjunctive support divided by its disjunctive support.

$$\text{bond}(X) = \text{sup}(X) / \text{dsup}(X)$$

Example: $\text{bond}(\{\text{pasta}, \text{orange}\}) = 3 / 4 = 0.75$

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange, cake}

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

□ The bond is a value in the $[0,1]$ interval

- 0 = no correlation
- 1 = maximum correlation

□ Examples:

- $\text{bond}(\{\text{cake}, \text{bread}\}) = 0 / 3 = 0$
- $\text{bond}(\{\text{lemon}, \text{bread}\}) = 1 / 3 = 0.33$
- $\text{bond}(\{\text{lemon}, \text{orange}\}) = 2 / 4 = 0.5$
- $\text{bond}(\{\text{pasta}, \text{lemon}\}) = 3 / 4 = 0.75$

Property 1 of the bond: The bond of an itemset X containing a single item is $\text{bond}(X) = 1$.

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

- Example: $\text{bond}(\text{bread}) = 1 / 1 = 1$
- This can be proven easily.

Property 2 of the bond (Apriori property): Let there be two itemsets X and Y . If $X \subset Y$, the bond of Y is less than or equal to the bond of X .

Example:

- The bond of {pasta} is 1
- The bond of {pasta, lemon} is $3 / 4 = 0.75$
- The bond of {pasta, lemon, orange} is $2 / 4 = 0.5$

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange, cake}

(bond is anti-monotonic)

Finding correlated frequent itemsets (using the bond)

□ Input:

- a transaction database
- a *minsup* threshold
- a *minbond* threshold $\in [0,1]$

□ Output:

- Each frequent itemset X such that:
 - $\text{sup}(X) \geq \text{minsup}$
 - $\text{bond}(X) \geq \text{minbond}$

Example

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

For *minsup* = 2 and *minbond* = 0.75,
the correlated frequent itemsets are:

{lemon}, {pasta}, {orange}, {cake}	bond = 1
{lemon, pasta},	bond = 0.75
{lemon, orange},	bond = 0.5
{pasta, orange},	bond = 0.75
{pasta, cake},	bond = 0.5
{orange, cake},	bond = 0.66
{lemon, pasta, orange}	bond = 0.5

Finding rare correlated frequent itemsets using the bond

□ Input:

- a transaction database
- a maxsup threshold
- a minbond threshold $\in [0,1]$

□ Output:

- Each rare itemset X such that:
 - $\text{sup}(X) < \text{maxsup}$
 - $\text{bond}(X) \geq \text{minbond}$

Example

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

For *maxsup* = 3 and *minbond* = 0.6,
the rare correlated itemsets are:

{bread} support = 1 bond = 1

{cake} support = 2 bond = 1

{orange, cake} support = 2 bond = 0.66

The CORI algorithm

- ❑ Based on **Eclat**
- ❑ The bond of all single items is 1.
- ❑ To find larger itemsets, **Eclat** combines pairs of itemsets X and Y to obtain a new itemset $Z = X \cup Y$
- ❑ To calculate the bond of Z , we need to find:
 - its conjunctive support
 - its disjunctive support.

CORI: Bouasker, S., Ben Yahia, S.: Key correlation mining by simultaneous monotone and anti-monotone constraints checking. In: Proc. 30th Symp. on Applied Computing, pp. 851-856, 2015.

Two structures

Each itemset X has two structures

- **TID-List** (Transaction-ID list):
list of transactions containing X .
- **DTID-List** (Disjunctive Transaction-ID list): list of transactions containing
at least one item from X .

Example →

TID-List and DTID-List

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

$TIDLIST(\{pasta\}) = \{T1, T2, T3, T4\}$

$TIDLIST(\{pasta, lemon\}) = \{T1, T2, T4\}$

$TIDLIST(\{bread, orange\}) = \{T1\}$

TID-List and DTID-List

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

$TIDLIST(\{pasta\}) = \{T1, T2, T3, T4\}$

$TIDLIST(\{pasta, lemon\}) = \{T1, T2, T4\}$

$TIDLIST(\{bread, orange\}) = \{T1\}$

$DTIDLIST(\{pasta\}) = \{T1, T2, T3, T4\}$

$DTIDLIST(\{pasta, lemon\}) = \{T1, T2, T3, T4\}$

$DTIDLIST(\{bread, orange\}) = \{T1, T3, T4\}$

TID-List and DTID-List

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

$TIDLIST(\{pasta\}) = \{T1, T2, T3, T4\}$ $sup(...) = 4$

$TIDLIST(\{pasta, lemon\}) = \{T1, T2, T4\}$ $sup(...) = 3$

$TIDLIST(\{bread, orange\}) = \{T1\}$ $sup(...) = 1$

$DTIDLIST(\{pasta\}) = \{T1, T2, T3, T4\}$

$DTIDLIST(\{pasta, lemon\}) = \{T1, T2, T3, T4\}$

$DTIDLIST(\{bread, orange\}) = \{T1, T3, T4\}$

TID-List and DTID-List

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

$TIDLIST(\{pasta\}) = \{T1, T2, T3, T4\}$

$sup(...) = 4$

$TIDLIST(\{pasta, lemon\}) = \{T1, T2, T4\}$

$sup(...) = 3$

$TIDLIST(\{bread, orange\}) = \{T1\}$

$sup(...) = 1$

$DTIDLIST(\{pasta\}) = \{T1, T2, T3, T4\}$

$dsup(...) = 4$

$DTIDLIST(\{pasta, lemon\}) = \{T1, T2, T3, T4\}$

$dsup(...) = 4$

$DTIDLIST(\{bread, orange\}) = \{T1, T3, T4\}$

$dsup(...) = 3$

If we have these lists, we can calculate the bond!

The CORI algorithm

- The TID-Lists and DTID-Lists of single items are calculated by reading the database.
- For any itemset $Z = X \cup Y$ that is obtained by joining two itemsets X and Y :
 - $TIDLIST(Z) = TIDLIST(X) \cap TIDLIST(Y)$
 - $DTIDLIST(Z) = DTIDLIST(X) \cup DTIDLIST(Y)$
 - $bond(Z) = |TIDLIST(Z)| / |DTIDLIST(Z)|$
- If $bond(Z) < minbond$, we don't need to check supersets of Z .

The All-CONFIDENCE Measure

Omiecinski, E.R.: Alternative interest measures for mining associations in databases. IEEE Transactions on Knowledge and Data Engineering 15(1), pp. 57-69 (2003)

All-confidence

The **all-confidence** of an itemset X is :

$$allconf(X) = \frac{\sup(X)}{\operatorname{argmax}\{support(i) | \forall i \in X\}}$$

Values are in $[0,1]$

Example: $allconf(\{\text{pasta}, \text{orange}\}) = \frac{3}{4} = 0.75$

Transaction	Items appearing in the transaction
T1	{ pasta , lemon, bread, orange }
T2	{pasta, lemon}
T3	{ pasta , orange , cake}
T4	{ pasta , lemon, orange , cake}

All-confidence

The **all-confidence** of an itemset X is :

$$allconf(X) = \frac{\sup(X)}{\argmax\{support(i) | \forall i \in X\}}$$

Values are in $[0,1]$

Example: $allconf(\{\text{lemon, orange, cake}\}) = \frac{1}{3} = 0.33$

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange, cake }

Property 1 of the all-confidence: The all-confidence of an itemset X containing a single item is $allconf(X) = 1$.

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange cake}

- Example: $allconf(cake) = 2 / 2 = 1$
- This can be proven easily.

Property 2 of the all-confidence (Apriori property): Let there be two itemsets X and Y . If $X \subset Y$, the all-confidence of Y is less than or equal to the all-confidence of X .

Example:

- The all-confidence of {pasta} is 1
- The all-confidence of {pasta, lemon} is $3 / 4 = 0.75$
- The all-confidence of {pasta, lemon, orange} is $2 / 4 = 0.5$

Transaction	Items appearing in the transaction
T1	{pasta, lemon, bread, orange}
T2	{pasta, lemon}
T3	{pasta, orange, cake}
T4	{pasta, lemon, orange, cake}

(all-confidence is anti-monotonic)

Algorithm

- ❑ It is easy to modify a frequent itemset mining algorithm such as **Apriori** and **Eclat** to calculate the **all-confidence**.
- ❑ We just need to pre-calculate the support of each single item.
- ❑ This is all we need to calculate the all-confidence.

Other correlation measures

Other correlation measures

- **Frequency affinity.**

Ahmed, C. F., Tanbeer, S. K., Jeong, B. S., Choi, H. J.: A framework for mining interesting high utility patterns with a strong frequency affinity. *Information Sciences*. 181(21), pp. 4878–4894 (2011)

- **Coherence**

Barsky, M., Kim, S., Weninger, T., Han, J.: Mining flipping correlations from large datasets with taxonomies. In: *Proc. 38th Int. Conf. on Very Large Databases*, pp. 370–381 (2012)

- **χ^2 , lift, cosine, kulc, maxconf**

Wu et al.: Re-examination of interestingness measures in pattern mining: a unified framework. *DMKD* 21:371-397 (2010)

- ...

Other correlation measures

Measure	Definition	Range	Null-invariant
$\chi^2(a, b)$	$\sum_{i,j=0,1} \frac{(e(a_i, b_j) - o(a_i, b_j))^2}{e(a_i, b_j)}$	$[0, \infty]$	No
$Lift(a, b)$	$\frac{P(ab)}{P(a)P(b)}$	$[0, \infty]$	No
$AllConf(a, b)$	$\frac{sup(ab)}{\max\{sup(a), sup(b)\}}$	$[0, 1]$	Yes
$Coherence(a, b)$	$\frac{sup(ab)}{sup(a) + sup(b) - sup(ab)}$	$[0, 1]$	Yes
$Cosine(a, b)$	$\frac{sup(ab)}{\sqrt{sup(a)sup(b)}}$	$[0, 1]$	Yes
$Kulc(a, b)$	$\frac{sup(ab)}{2} \left(\frac{1}{sup(a)} + \frac{1}{sup(b)} \right)$	$[0, 1]$	Yes
$MaxConf(a, b)$	$\max \left\{ \frac{sup(ab)}{sup(a)}, \frac{sup(ab)}{sup(b)} \right\}$	$[0, 1]$	Yes

Note: the definitions are given the case of two items **a** and **b**.

Conclusion

- The problem of **rare itemset mining**
- **Three definitions of rare itemsets:**
 - Infrequent itemsets
 - Minimal rare itemsets
 - Perfectly rare itemsets
- **Two algorithms:**
 - AprioriRare
 - AprioriInverse
- **correlated patterns**
 - bond
 - all-confidence
 - other correlation measures
 - the CORI algorithm